

Literature Benchmark Brief: Ela_2011_PESGM

Title: Operating reserves and variable generation: A survey of international practices

Authors: E. Ela, M. Milligan, B. Kirby (2011)

Venue: IEEE PES General Meeting | **DOI:** 10.1109/PES.2011.6039832

Cluster: Cluster D: Probabilistic Sizing & Reserve

Core Theoretical Mechanism & Model Formulation

Survey of operating reserve definitions: regulating, contingency, and ramping reserves

Empirical Scale & Experimental Dataset

North American and European ISO/RTO operational frameworks

Relevance to IEEE TSTE & Sustainable Energy Systems

Clarifies terminology between real-time dispatch, regulating reserves, and contingency margins

Identified Gaps & Operational Vulnerabilities

Industry survey without mathematical algorithmic optimization

Role in Manuscript Defense & Reviewer #2 Proof

Defends replacement of AGC with Real-Time Economic Dispatch (RTED) at 10-min resolution